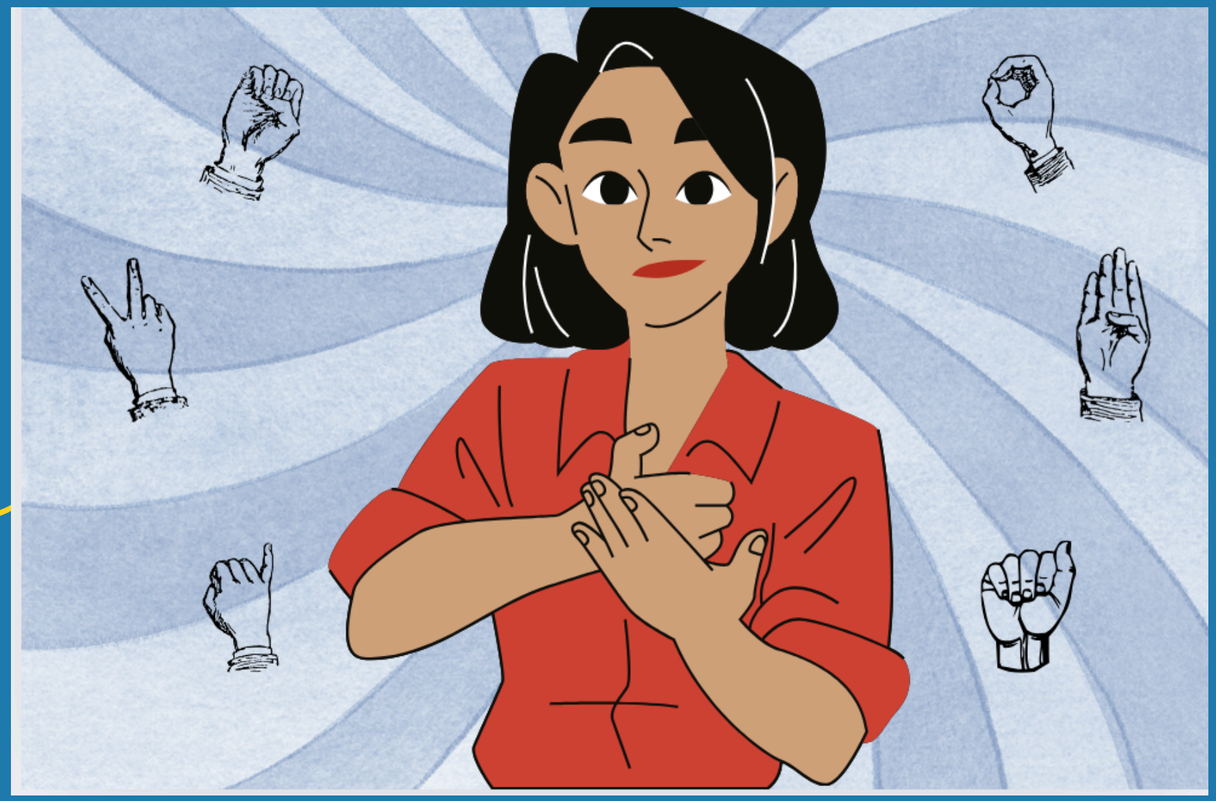
SIGN LANGUAGE TO VOICE GENERATION







Sign language is a vital communication tool for people who are deaf or hard of hearing, but not everyone understands it. This project focuses on converting sign language gestures into spoken words using image processing and machine learning. By recognizing sign language from images captured by a camera, the system generates corresponding voice output. This approach aims to bridge the communication gap, making interactions more inclusive and accessible for everyone.

PROBLEM STATEMENT

The problem this project solves is the communication gap between people who use sign language and those who don’t understand it. Many deaf or hard-of-hearing individuals face challenges when interacting with others who cannot sign. This project aims to create a system that converts sign language into spoken words, making communication easier and more inclusive.





RESEARCH PAPER

https://www.sciencedirect.com/science/article/abs/pii/S2214785320350343

https://arxiv.org/pdf/1811.11997

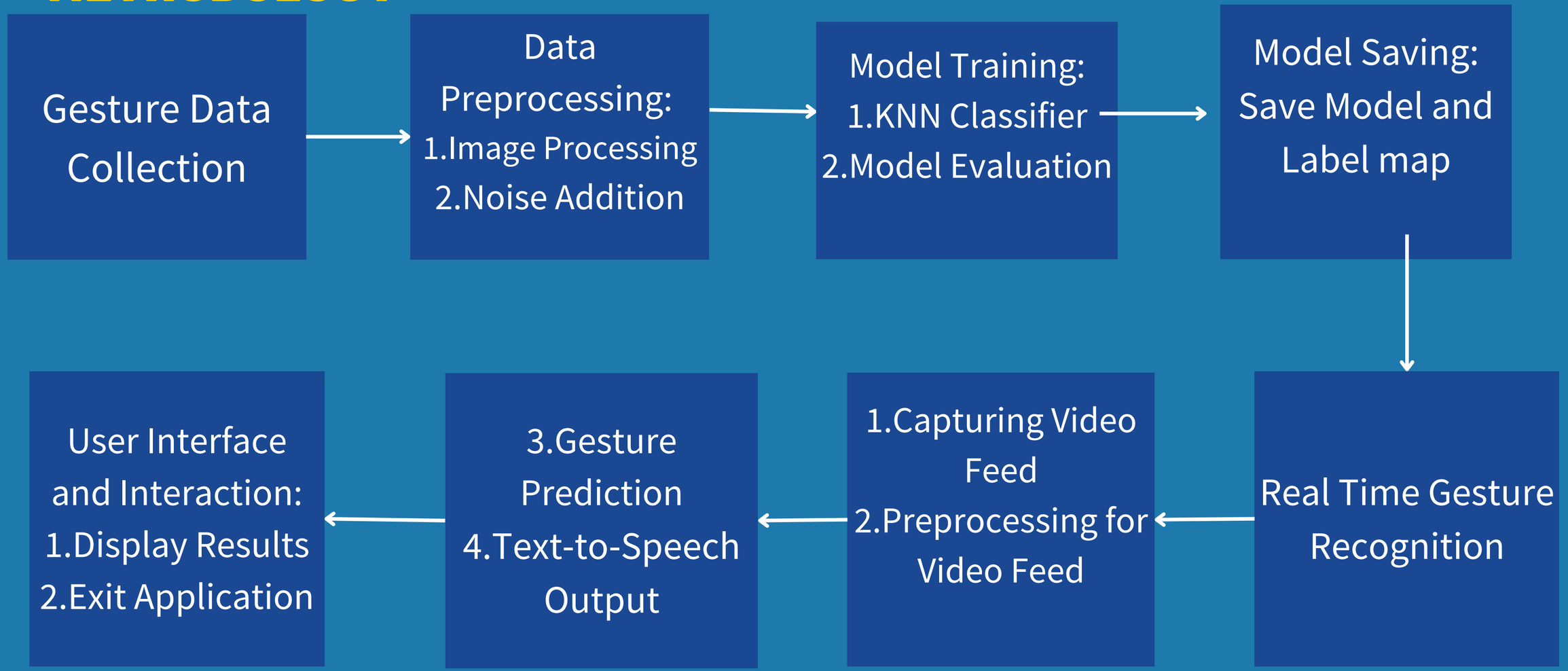
https://ieeexplore.ieee.org/document/9743064



DATASET LINK	**https://www.kaggle.com/datasets/prathumarikeri/american-sign-language-09az**



METHODOLOGY





MODEL ACCURACY : The trained KNN model achieved 100% accuracy on the test dataset, demonstrating its effectiveness in recognizing gestures from the preprocessed images.

SYSTEM PERFORMANCE: During real-time testing, the system successfully detected sign language gestures from the webcam feed and converted them into spoken words using the text-to-speech engine.

APPLICATION AND BENEFITS:

- Education , Public Services , Daily Interactions , Communication Aid

FUTURE SCOPE :

1. Multilingual Support
2. Real Time Translation
3. Integration with Apps
4. Smart Device Integration
5. Customizable Voice Output

LIBRARIES:

- 1. Image Processing & Video Capture
- OpenCV (cv2): For capturing video from the webcam and processing images in real-time.
- 2. Numerical Computing
- NumPy (NumPy): Efficiently handles image data as arrays and applies transformations like noise.
- 3. Machine Learning
- scikit-learn (Sk learn):
- KNN Classifier : Gesture classification model.
- train test split : Splits the dataset into training and testing sets.
- accuracy score : Measures the accuracy of the model.
- 4. Model Persistence
- Joblib : Used to load the pre-trained gesture recognition model and label mapping.
- 5. Text-to-Speech Conversion
- pyttsx3 : Converts recognized gestures into voice output.
- 6. Utility & System Libraries
- os: Manages file directories.
- matplotlib

CONCLUSION:

Effective Communication Bridge: The system converts ASL gestures into voice, enhancing communication between sign language users and non-users.

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- Real-Time Gesture Recognition :Successfully implemented real-time recognition using a webcam, ensuring quick and efficient interaction.
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- High Accuracy and Robustness: The model performed well even in noisy environments due to effective preprocessing and noise-handling techniques
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